

Question description

Question No:8

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
5 6
2 5 4 5 2
5 3
1 6 3 5 2
```

EXPECTED OUTPUT

```
-1
3
```

Test Case 2

INPUT (STDIN)

```
2
5 6
12 15 14 15 12
5 3
11 61 31 51 21
```

EXPECTED OUTPUT

```
9
58
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
void sort(int a[],int n)
```

Test Case 2

KEYWORD

```
for(i=0;i<n-1;i++)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

5

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

40

Code :

```
1  #include <stdio.h>
2  int main() {
3      int T;
4      scanf("%d", &T);
5
6      while (T--) {
7          int N, K;
8          scanf("%d %d", &N, &K);
9
10         int max = 0;
11
12         for (int i = 0; i < N; i++) {
13             int x;
14             scanf("%d", &x);
15
16             if (x > max) {
17                 max = x;
18             }
19         }
20
21         if (max <= K)
22             printf("-1\n");
23         else
24             printf("%d\n", max - K);
25     }
26     return 0;
27 }
```

Output :

Test Case-1 :

	2	
	5 6	
	2 5 4 5 2	
	-1	
	5 3	
	1 6 3 5 2	
	3	

Test Case-2 :

	2	
	5 6	
	12 15 14 15 12	
	9	
	5 3	
	11 61 31 51 21	
	58	

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